

Developing a Feasible Neuroplastic Mechanical Neuron in the Domain of Soft Robotics

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Abstract

Onboard computation remains one of the open problems of soft robotics. This article develops a mechanically feasible neuron that can serve as the building block of a neuroplastic mechanical neural network. Taking the organisation of the biological neuron as inspiration, the proposed neuron encodes information in fluid pressure: input pressures act on flexible membranes that bound a sealed activation chamber, so that the inputs are integrated into a single field-based activation pressure without being consumed, and a pressure-controlled pneumatic valve converts this activation pressure into a nonlinear output signal. The equilibrium of the membranes shows that the activation pressure is a compliance-weighted combination of the input pressures, in which the mechanical weight of each input is the ratio of its membrane area to its volumetric stiffness. Two mechanisms for changing these weights after fabrication are proposed: limiting the deformation path of the membrane, and tuning the bulk modulus of an intermediate fluid between two membranes. The sensitivity of the activation pressure to the chamber compliance, the weight membranes, a bias membrane and the input pressures is derived. When the membrane stiffnesses are constant, the neuron is exactly an artificial neuron, a weighted sum with a bias followed by a sigmoid, with weights that are non-negative and, together with the bias and chamber weights, sum to one; the reachable hyperplanes therefore all contain the equal-pressure diagonal, whose crossing point is set by the bias pressure. With deformation-dependent stiffness, the neuron becomes a nonlinear, state-dependent computational element of which the artificial neuron is a special case.

1 Introduction

1.1 Background

One of the remaining frontiers in soft robotics is the development of robust onboard computation and control. A compelling long-term goal in this field is the creation of a fully mechanical robot capable of intelligent agency: a system that can sense its environment, interact with it, and adapt autonomously based on those interactions [1].

Nature has already achieved this through neural systems. Although biological organisms contain many forms of computation, the brain remains the most sophisticated example of an integrated computational kernel. It is the central neural network that underlies animal agency, and it is built from neural networks composed of simple computational units known as neurons. In isolation, a single neuron is a modest computational element, but when many neurons are interconnected, complex behaviour emerges. This broader principle, where intricate global structure arises from the interaction of simple local rules, is widely observed in nature. Stephen Wolfram has similarly speculated that physical reality itself may arise from repeated application of simple computational rules, as developed in *A New Kind of Science* [2].

The human brain contains more than 100 billion neurons, with each neuron potentially connecting to up to 10 000 other neurons and passing signals through as many as 100 trillion synaptic connections [3, 4]. This vast network of interactions is one of the reasons the human brain exhibits such remarkable computational capacity.

For this reason, there is a strong need to develop a mechanical neuron that can serve as a foundational building block for mechanical neural networks (MNNs). Such a device would not merely mimic neural structure superficially; it would need to support reprogrammable behaviour so that the resulting MNN could continue learning, adapting, and reorganising in response to its environment. A true neuroplastic mechanical neuron would therefore provide the basis for autonomous mechanical systems that do not simply react but evolve their responses over time.

1.2 Objective and scope

The objective of this work is to develop and characterise a mechanically feasible neuron that can serve as a foundational building block for future neuroplastic mechanical neural networks. The proposed neuron is developed from the physical organisation and functional roles of a biological neuron, with pressure-based mechanical systems used to reproduce the processes of signal reception, integration, activation, and transmission. The intention is therefore not to reproduce a biological neuron in its entirety, but to use its physical organisation as inspiration for a mechanically feasible computational architecture.

The mechanical behaviour of the proposed neuron is subsequently formulated mathematically to establish how its physical parameters determine its computational response. In particular, the analysis considers the mechanical integration of multiple input pressures, the influence of membrane stiffness and geometry on the resulting activation pressure, and the conversion of this activation into a nonlinear output. The ability to alter the mechanical parameters responsible for input weighting is also investigated, providing a physical basis for modifying the computational behaviour of the neuron after fabrication.

The resulting mechanical formulation is further related to the conventional artificial neuron. When the relevant membrane stiffnesses are assumed to remain constant through deformation, the mechanical equilibrium can be rearranged into a weighted-sum form with a bias term, and the nonlinear output relationship gives the corresponding artificial-neuron formulation. This result provides a direct mathematical interpretation of the proposed mechanical neuron in terms of an artificial neural-network neuron. However, constant stiffness represents a restricted case of the more general mechanical system. Since membrane stiffness can vary with deformation, the effective mechanical weights may themselves depend on the state of the neuron. The general formulation therefore describes a nonlinear, state-dependent neural computation element, of which the conventional artificial neuron is a special case.

Finally, the mechanically adjustable parameters of the proposed architecture are considered in the context of future neuroplasticity. This work does not implement an autonomous learning rule or demonstrate a complete neuroplastic mechanical neural network. Rather, it establishes the mechanically reconfigurable neuron required as a physical foundation for such a system. The development therefore progresses from biological inspiration to a mechanically feasible architecture, from mechanical behaviour to its mathematical computational representation, and finally to the reconfigurability required for future neuroplastic mechanical computation.

2 The neuron

2.1 Neuron anatomy

This study focuses on the standard biological neuron and its major components, although alternative neuron topologies exist. As illustrated in Fig. 1, a typical neuron consists of a soma, dendrites, a single axon, and synapses [3]. While signal transmission in biological neurons is electrochemical in nature, this work primarily focuses on the computational aspect of information processing (the kernel) rather than the underlying mechanics.

Inputs to a neuron are received by the dendrites from other neurons. The dendrites process and integrate these inputs through local computations [3, 4], which may involve complex

Neuron Anatomy

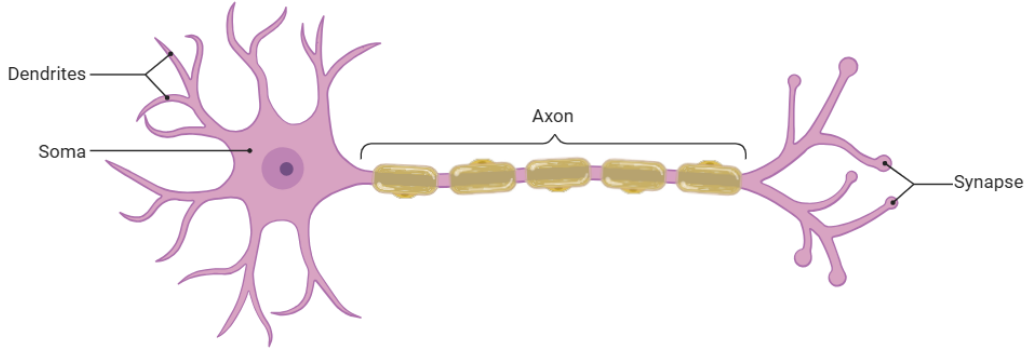


Figure 1: Anatomy of a biological neuron. The neuron consists of dendrites, which receive input signals from other neurons; the soma (cell body), where inputs are integrated; the axon, which transmits electrical signals to other neurons; and the synapse, where the axon communicates with the dendrites or soma of neighbouring neurons through chemical or electrical signals.

interdependent nonlinear interactions [5, 6, 7, 8]. These interactions are integrated into a single value in the soma, which determines the neuron’s response. When this integrated signal exceeds a certain threshold, the neuron generates an action potential, a binary-like signal corresponding to a “fire” or “no-fire” state. This action potential then propagates along the axon to the synapses, which interface with the dendrites of downstream neurons, allowing the signal to traverse the network [3, 4].

Although the magnitude of a neuron’s action potential is effectively binary, the information it conveys is not necessarily so. Neurons can encode information through temporal dynamics, most notably rate coding, in which signal intensity is represented by the frequency of action potentials rather than their amplitude [8, 9, 10]. This type of neural signalling is analogous to pulse-width or pulse-frequency modulation: the information is not encoded in the spike magnitude but in the firing rate or spike timing. As a result, rich, continuous information processing emerges from sequences of discrete, all-or-nothing events.

From this perspective, the fundamental operations of a neuron can be conceptualised as a sequence of systematic stages: input reception via synaptic–dendritic connections; dimensional collapse through inter-dendritic computation into a unified neuronal activation within the soma; propagation of the resulting action potential along the axon; and finally, transmission of this signal through synaptic connections to downstream neurons.

Accordingly, a neuron can be viewed as a system that receives multiple input signals, combines them into a single activation through a parametric, potentially interdependent nonlinear relationship, and transmits the resulting activation signal to the inputs of other neurons.

2.2 Biological neural signal transmission

Biological neurons transmit information through two complementary physical substrates, transport and field, that operate in sequence across neuronal compartments. In dendrites, signalling is predominantly transport-based: synaptic inputs open ion channels and generate local ionic currents that propagate through the cytoplasm toward the soma, so information is conveyed by the movement of charge along dendritic cable-like structures [11].

As these currents reach the soma, the neuron shifts into a field-based regime of integration. In this regime, changes in membrane potential regulate voltage-gated channels and conductances, enabling the cell to combine multiple inputs through a local electrical state rather than through long-range ionic transport across the entire compartment [12, 13]. This is the crucial computational step: the neuron does not simply pass the input through unchanged but converts it into

an internal state that determines whether and how an output will be generated.

At chemical synapses, the process reverses. Depolarisation of the presynaptic membrane triggers vesicle fusion and neurotransmitter release, which then diffuses across the cleft and produces transport-based ionic currents in the postsynaptic cell [14]. In this way, the neuron transforms an incoming transport-based signal into a field-mediated internal response, and then back into a transport-based output.

This separation of transport-based sensing, field-mediated integration, and transport-based output is especially important for mechanical neuron design because it enables non-destructive signal processing. The input is not consumed by the act of computation; rather, it is sampled and transformed into a local state that can drive an output response while preserving the integrity of the upstream signalling pathway. For artificial systems, this distinction addresses a central engineering challenge: how to build a neuron that can compute on incoming information without exhausting or destroying the signal that carries it.

2.3 Neural plasticity

Neural plasticity refers to the ability of a neuron to adapt by updating its internal parameters. In biological neural networks, plasticity is a persistent and ongoing process: synaptic strengths are continuously modified as organisms learn, adapt, and refine internal models of the world. This continual adaptation allows biological systems to remain responsive to changing environments rather than converging to a fixed functional state [15, 16].

MNNs can be broadly classified according to their degree of neural plasticity. The first class comprises non-plastic MNNs, in which the network parameters are determined prior to fabrication, often through analytical design, numerical optimisation, or simulation, and are immutable once the system is manufactured. Such systems perform computation through their mechanical structure but lack the ability to adapt or learn after deployment.

The second class consists of neuroplastic MNNs, in which network parameters can be modified after fabrication. Within this class, further distinctions can be made based on the mechanism by which plasticity is achieved. Mechanically neuroplastic MNNs implement parameter updates entirely through mechanical processes, enabling adaptation without external domain intervention; the mechanical kernel itself is then solely responsible for the parameter updating. In contrast, hybrid neuroplastic MNNs retain a mechanically embodied network structure but rely on external systems to compute and apply parameter updates.

Representative examples of hybrid neuroplastic MNNs include architected materials that learn behaviours [17], in which the mechanical network structure is adapted through electromechanical actuation, and *Beyond Silicon* [18], where parameter updates to the mechanical network are performed manually.

3 Development of a mechanically feasible neuron

This section develops a conceptual mechanically feasible neuron by first defining the admissible design space and the functional requirements imposed on the physical system. The design is initially formulated in an implementation-agnostic manner, after which a topology inspired by the biological neuron is introduced. The resulting model is then characterised mathematically to determine how its parameters influence signal transmission, integration, thresholding, and plasticity.

3.1 Design-space definition

Before considering the physical structure of a mechanical neuron, it is necessary to define its computational behaviour. Rather than prescribing a specific mechanical implementation, this definition establishes the minimum computational requirements any proposed neuron must

satisfy. Because signal processing in embodied systems relies directly on physical mechanisms, computational and physical constraints are intrinsically interconnected.

The neuron is treated as a computational unit that receives multiple inputs, combines them according to a parameter-dependent rule, and produces an output signal. A given neuron i receives an input vector

$$\vec{x}_i \in \mathbb{R}^{n_i}, \quad (3.1)$$

where n_i is the number of incoming signals. These inputs are received as transport-based signals, which are transformed into a field-based scalar activation signal according to

$$z_i = \sigma_i(f_i(\vec{x}_i, \boldsymbol{\theta}_i)) \in \mathbb{R}. \quad (3.2)$$

In this expression, $f_i(\cdot, \boldsymbol{\theta}_i)$ represents the internal input-processing operation of the neuron. It maps the n_i -dimensional transport-based input signal vector to a single field-based scalar quantity. The parameter vector

$$\boldsymbol{\theta}_i \in \Theta \subset \mathbb{R}^p \quad (3.3)$$

contains the physical parameters that determine the behaviour of the neuron, which must be dynamically changeable or reconfigurable to support plasticity. The function $\sigma_i(\cdot)$ represents the activation operation, which maps the field-based scalar activation signal into a transport-based output signal via a nonlinear mapping to be transmitted downstream.

3.2 Biologically inspired mechanical neuron topology

3.2.1 Information encoding and signalling

Computational information within the mechanical neuron is encoded in fluid pressure, with kernel operations performed through the manipulation of pressure states. In an enclosed fluidic system, pressure is transmitted through the substrate as a pressure field and does not require continuous fluid flow. This represents a field-based signalling substrate, in which the pressure state is determined by the conditions throughout the enclosed fluid volume. In a flow-based system, by contrast, the pressure depends on the dynamics of fluid moving through the substrate. The signal is therefore transmitted through the transport of fluid along a channel and represents a transport-based signalling substrate.

3.2.2 Input-to-activation kernel

3.2.2.1 Conceptual design of the flexible-membrane input interface

The input-to-activation stage of the neuron is implemented using flexible membranes that separate the input fluidic channels from a central activation chamber. The activation chamber is filled with a confined fluid, while each input channel conveys an externally generated pressure signal. Pressure applied within an input channel causes its corresponding membrane to deform, thereby transmitting a force to the fluid contained in the central chamber. The contributions from the individual input pressures are consequently combined through the mechanical interaction of the membranes and the activation fluid. The resulting pressure within the central chamber constitutes the neuron's scalar activation signal.

Figure 2 illustrates a two-input configuration of this mechanism. Changes in the input pressures alter the forces acting on the respective membranes, which in turn modify the pressure of the fluid contained within the activation chamber. The chamber pressure therefore provides a single internal state that reflects the combined influence of the input signals.

This configuration also provides a natural means of achieving non-destructive signal processing. The input signals are supplied by pressure sources operating at a prescribed steady-state pressure with continuous flow through the input channels. In contrast, the fluid within the activation chamber is enclosed and does not flow into or out of the chamber during operation.

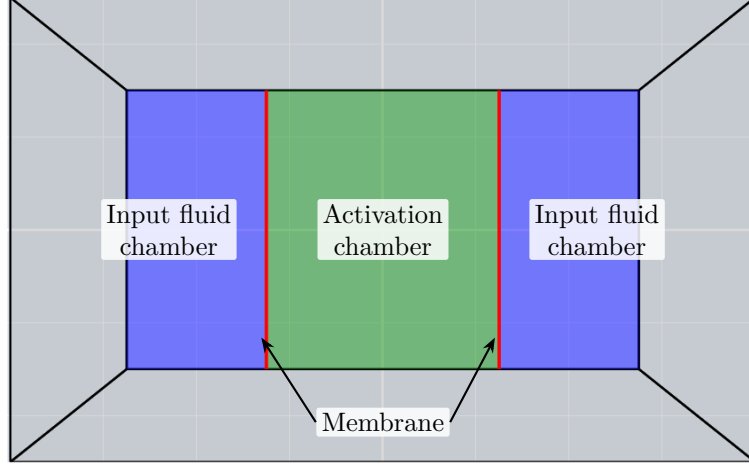


Figure 2: Two-input configuration of the flexible-membrane input interface. Each input channel is separated from the central activation chamber by a flexible membrane.

The input pressures therefore influence the activation state through membrane-mediated force transmission, rather than through the transfer of fluid into the activation chamber. The input signals can consequently be processed without being consumed or directly altered by the activation mechanism.

The proposed arrangement thus implements the first stage of the mechanical neuron’s computational operation: multiple transport-based input signals are converted into a single field-based activation signal.

3.2.2.2 Mathematical model of the flexible-membrane input interface

The following characterises the mechanical behaviour of the membranes and activation chamber and establishes how the input pressures determine the resulting chamber pressure.

At equilibrium, each flexible membrane deforms until the elastic restoring force generated by its deformation balances the net pressure force acting across it. The pressure difference across a membrane is determined by the pressure in the corresponding input channel and the pressure in the central activation chamber.

For membrane j , let p_j denote the pressure in input channel j , p_a the pressure in the activation chamber, and p_{atm} the atmospheric pressure. If A_j is the effective area of the membrane and F_j^e is its elastic restoring force, the equilibrium condition is

$$F_j^e = (p_j - p_a) A_j. \quad (3.4)$$

This expression assumes a sign convention in which positive membrane deformation is produced by an input pressure greater than the activation-chamber pressure. More generally, the membrane equilibrium may be written as

$$F_j^e(\delta_j) - (p_j - p_a) A_j = 0, \quad (3.5)$$

where δ_j is the equilibrium deformation of membrane j . The elastic force is therefore a function of deformation, and the deformation is determined implicitly by the pressure difference across the membrane.

To relate membrane deformation to its restoring force, the deformation is represented in terms of the volume displaced by the membrane rather than its local displacement field. Let ΔV_j denote the volumetric deformation of membrane j , defined as the change in activation-chamber volume produced by its deformation. A volumetric stiffness K_j can then be introduced to describe the resistance of the membrane to this deformation. The elastic restoring force is expressed as

$$F_j^e = K_j(\cdot) \Delta V_j. \quad (3.6)$$

This represents a generalised stiffness relation, in which the stiffness may vary with the current volumetric deformation. Substituting it into the membrane equilibrium gives

$$K_j(\cdot) \Delta V_j = (p_j - p_a) A_j, \quad (3.7a)$$

$$\Delta V_j = \frac{(p_j - p_a) A_j}{K_j(\cdot)}. \quad (3.7b)$$

Under the assumption that the activation fluid is incompressible and the chamber is sealed, the total change in fluid volume must be zero. Including the m input membranes, the condition is

$$\sum_{j=1}^m \Delta V_j = 0. \quad (3.8)$$

Substituting the membrane deformation relations gives

$$\sum_{j=1}^m \frac{(p_j - p_a) A_j}{K_j(\cdot)} = 0. \quad (3.9)$$

For constant effective stiffnesses, solving for the activation-chamber pressure yields

$$p_a = \frac{\sum_{j=1}^m \frac{A_j}{K_j(\cdot)} p_j}{\sum_{j=1}^m \frac{A_j}{K_j(\cdot)}}. \quad (3.10)$$

Thus, the equilibrium activation pressure is a stiffness- and area-weighted combination of the input pressures. The weighting of each input is determined by its membrane's effective compliance, $A_j/K_j(\cdot)$. Since this quantity determines the relative contribution of each input pressure to the activation pressure, it is analogous to a weight in an artificial neural network. The effective mechanical weight of membrane j is therefore defined as

$$W_j(\cdot) = \frac{A_j}{K_j(\cdot)}. \quad (3.11)$$

Using this definition, the activation pressure becomes

$$p_a = \frac{\sum_{j=1}^m W_j(\cdot) p_j}{\sum_{j=1}^m W_j(\cdot)}. \quad (3.12)$$

Since $W_j(\cdot) > 0$, the activation pressure is a convex combination of the input pressures.

3.2.3 Mechanically reconfigurable weights

3.2.3.1 Weight-modification principle

The mechanical weight associated with each input must be physically adjustable for the neuron to exhibit neuroplastic behaviour. Since the weight is defined by the ratio between the effective membrane area and stiffness, it can be modified by changing either of these properties. The proposed approach focuses on modifying the effective stiffness through controlled deformation of the input membrane.

The stiffness of membrane j is not treated as a fixed quantity. Instead, it is considered a function of the membrane's equilibrium deformation. The deformation itself depends on the pressure difference across the membrane, although it may also be influenced by other physical, geometric, material, and boundary-condition variables. These dependencies are represented as

$$K_j = K_j(\delta_j, \cdot) \quad (3.13)$$

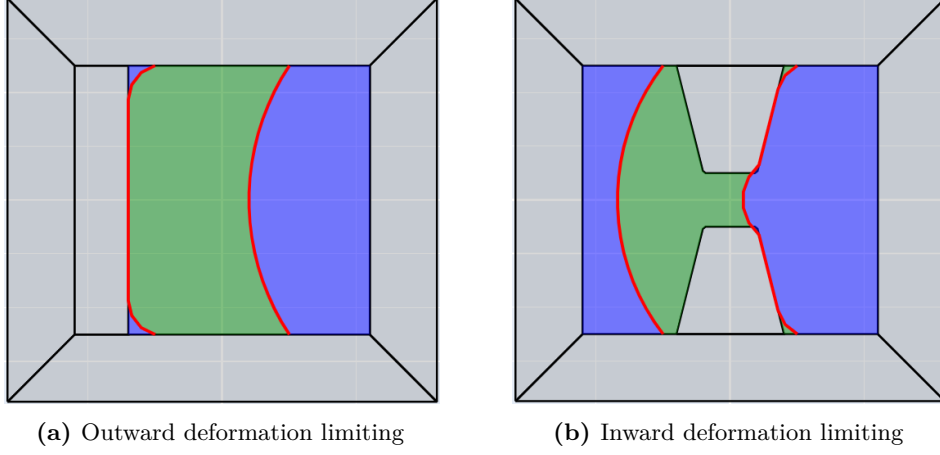


Figure 3: Methods for altering membrane deformation. (a) Outward membrane deformation restriction, limiting the membrane’s deformation away from the activation chamber. (b) Inward membrane deformation restriction, limiting the membrane’s displacement into the activation chamber.

and

$$\delta_j = \delta_j(p_j - p_a, \cdot), \quad (3.14)$$

where the notation $(\cdot)$ represents additional variables that may influence the respective relationship. Substituting these dependencies into the definition of the mechanical weight gives

$$W_j = \frac{A_j}{K_j(\delta_j(p_j - p_a, \cdot), \cdot)}. \quad (3.15)$$

The weight may therefore be represented more generally as

$$W_j = W_j(p_j - p_a, \cdot). \quad (3.16)$$

This relationship demonstrates that the mechanical weight is coupled to the equilibrium state of the membrane. By controlling the deformation of membrane j , its effective stiffness can be altered, and its corresponding weight can consequently be updated. The pressure difference across the membrane provides one means of influencing this deformation, but it is not the sole determinant of the weight. Rather, the weight depends on the complete mechanical state of the membrane and the other variables governing its deformation and stiffness.

3.2.3.2 Proposed mechanical realisation of the weight update

Deformation limiting. Deformation-limited weight tuning operates by restricting the membrane’s available deformation path. Since the membrane can deform in either direction, the limiting obstacle may be positioned either on the activation-chamber side, producing inward deformation limiting, or on the input-channel side, producing outward deformation limiting, as shown in Fig. 3.

By controlling the shape of the limiting obstacle, the allowable equilibrium deformation of the membrane can be controlled. Deformation can be limited independently in either direction, such that

$$\delta_j = \begin{cases} \delta_j^{\text{in}}, & p_a < p_j, \\ \delta_j^{\text{out}}, & p_a > p_j. \end{cases} \quad (3.17)$$

Here, δ_j^{in} and δ_j^{out} denote the inward and outward deformation limits, respectively. Since the effective stiffness is a function of the membrane deformation, $K_j = K_j(\delta_j, \cdot)$, the mechanical weight is consequently also affected:

$$W_j = \frac{A_j}{K_j(\delta_j, \cdot)}. \quad (3.18)$$

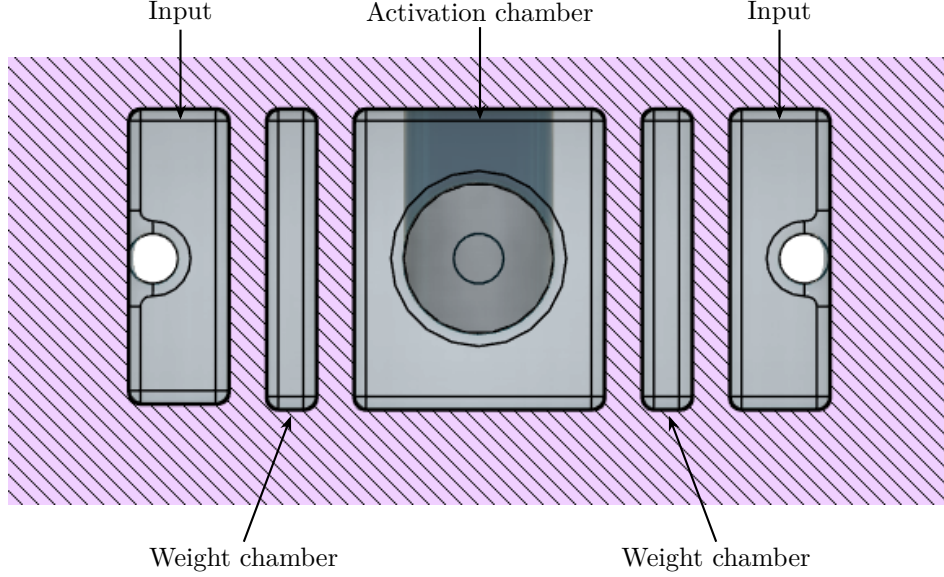


Figure 4: Conceptual schematic of an intermediate-fluid bulk-modulus-tuned input-weight kernel: flexible membranes separated by an intermediate (weight) chamber whose tuneable compressibility controls the transmission of deformation and force from the input-pressure chamber to the activation-pressure chamber.

Deformation limiting therefore provides a mechanical means of updating the weight. Changing the deformation path changes the membrane’s equilibrium deformation and effective stiffness, which consequently changes the contribution of the corresponding input to the activation pressure.

Inward deformation limiting is expected to introduce an additional complication because the limiting obstacle would be located inside the activation chamber. Changing its volume changes the volume available to the confined activation fluid, coupling the weight-update mechanism to the chamber’s pressure. This complication is avoided when the deformation limit is implemented on the input-channel side.

Intermediate-fluid bulk-modulus tuning. Force-transfer limiting is implemented by dividing the original membrane into two flexible membranes with a confined fluid layer between them (Fig. 4). The input pressure acts on the outer membrane, while the activation-chamber pressure acts on the inner membrane. The pressure of the intermediate fluid, referred to as the weight pressure, transmits deformation between the two membranes.

This configuration introduces three pressure variables: the input pressure p_i , the activation-chamber pressure p_a , and the weight pressure p_w . Let ΔV_1 denote the deformation of membrane 1 into the intermediate chamber, with K_1 representing its volumetric stiffness. Similarly, let ΔV_2 denote the deformation of membrane 2 into the activation chamber, with K_2 representing its volumetric stiffness.

It is important to note that ΔV_j is defined as the change in the volume of the activation chamber. Consequently, K_j represents the volumetric stiffness associated with deformation into the activation chamber due to the difference between the input pressure p_i and the activation pressure p_a . The detailed derivation of F_j , K_j , and their dependence on the intermediate-fluid properties is provided in Appendix A.

The intermediate fluid provides the tuneable mechanical element. Its bulk modulus B determines its resistance to volumetric compression and therefore controls how effectively deformation is transferred from membrane 1 to membrane 2. A low bulk modulus allows the intermediate fluid to undergo a comparatively large volume change, reducing force transfer to membrane 2. In the limiting case $B = 0$, the intermediate fluid provides no resistance to compression, and the

effective stiffness of the combined mechanism approaches

$$K_j(B = 0) \rightarrow \infty. \quad (3.19)$$

In contrast, as the bulk modulus increases, the intermediate fluid becomes less compressible and transfers an increasing proportion of the deformation between the two membranes. In the limiting case $B \rightarrow \infty$, the intermediate fluid is incompressible, and the deformation of the two membranes becomes fully coupled. The effective stiffness then approaches the sum of the individual membrane stiffnesses:

$$K_j(B \rightarrow \infty) = K_1 + K_2. \quad (3.20)$$

The effective stiffness can therefore be represented as a function of the bulk modulus,

$$K_j = K_j(B), \quad (3.21)$$

with the limiting behaviour

$$K_j(0) = \infty, \quad K_j(\infty) = K_1 + K_2. \quad (3.22)$$

Since the mechanical weight is inversely related to the effective stiffness, it follows that

$$W_j(B) = \frac{A_j}{K_j(B)}. \quad (3.23)$$

The bulk modulus of the intermediate fluid therefore provides a physical means of tuning the mechanical weight. By changing B , the force-transfer capability of the two-membrane system is altered, which changes the effective stiffness K_j and consequently the influence of the corresponding input on the activation pressure. Bulk modulus is conventionally defined as the pressure change associated with volumetric strain, supporting this interpretation of B as the fluid's resistance to compression.

3.2.4 Activation-to-output kernel

The activation pressure p_a is a field-based internal state. To communicate this state to another mechanical neuron, the activation pressure must act on an output interface that generates a pressure-driven flow or pressure pulse in an external output channel.

The proposed design uses a pressure-controlled pneumatic switching interface to perform this conversion. The interface consists of four fundamental components: a pressure source, an atmospheric vent, a flow resistor, and a mechanically actuated valve. The pressure source establishes the high-pressure state of the output signal, while the atmospheric vent establishes the low-pressure state. The flow resistor restricts fluid flow. The mechanically actuated valve is controlled by the activation pressure p_a , opening or closing according to whether the activation state is low or high.

3.2.4.1 Inverted output configuration

The inverted output configuration is illustrated in Fig. 5a. In this configuration the pressure-controlled mechanical valve is on the pressure-source side, and the flow resistor is on the vent side. The output connection to the downstream neuron inputs is located between the flow resistor and the valve.

Assume the pressure valve is normally open. When the activation pressure p_a is low, the valve remains open and connects the output to the pressure source. Fluid continuously escapes through the flow resistor to atmosphere, but this loss is replenished by the pressure source. The output, which is connected to neuron inputs that do not draw fluid flow, therefore remains at a high pressure. When p_a is high, the valve closes and disconnects the output from the pressure

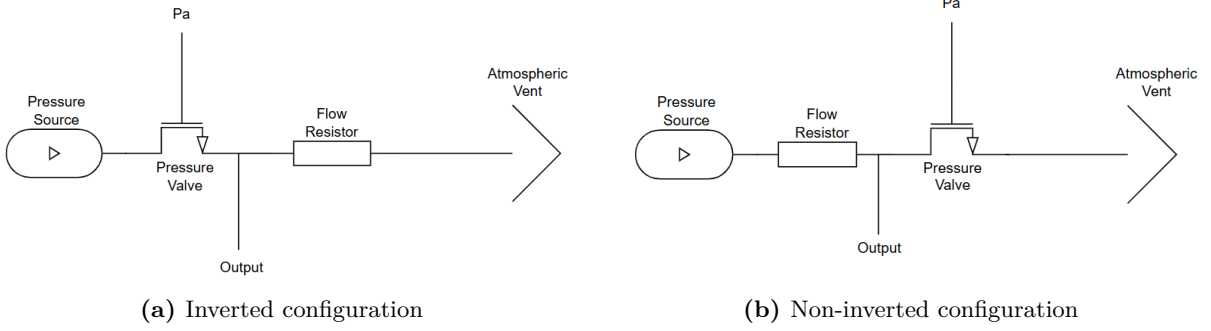


Figure 5: Activation-to-output configurations with a normally open, pressure-controlled mechanical valve driven by p_a . (a) Inverted: the valve sits between the pressure source and the output connection, and the output is connected through a flow resistor to atmosphere, so that a low activation pressure maintains a high output pressure and a high activation pressure produces a low output pressure. (b) Non-inverted: the flow resistor sits between the pressure source and the output connection, and the valve between the output and the atmospheric vent, so that a low activation pressure produces a low output pressure and a high activation pressure produces a high output pressure.

source. The output remains connected to atmosphere through the flow resistor, allowing fluid to escape without being replenished. The output pressure therefore decreases to the low-pressure state. Consequently, the configuration produces an inverted output relationship:

$$p_a \text{ low} \Rightarrow p_{\text{out}} \text{ high}, \quad p_a \text{ high} \Rightarrow p_{\text{out}} \text{ low}.$$

3.2.4.2 Non-inverted output configuration

The non-inverted output configuration is illustrated in Fig. 5b. In this configuration, the flow resistor is positioned on the pressure-source side, while the pressure-controlled mechanical valve is positioned on the vent side. The output connection to the downstream neuron inputs is located between the flow resistor and the valve.

Assuming again that the pressure valve is normally open, a low activation pressure p_a keeps the valve open and connects the output to atmosphere. The output pressure can therefore escape through the valve, while the pressure source replenishes the output through the flow resistor. Because the replenishment flow is restricted, the output pressure decreases to the low-pressure state. When p_a is high, the valve closes and disconnects the output from atmosphere. The output remains connected to the pressure source through the flow resistor, allowing its pressure to increase towards the source pressure. Once the valve is closed, the output pressure cannot escape through the vent and therefore remains at the high-pressure state. Consequently, the configuration produces a non-inverted output relationship:

$$p_a \text{ low} \Rightarrow p_{\text{out}} \text{ low}, \quad p_a \text{ high} \Rightarrow p_{\text{out}} \text{ high}.$$

4 Mathematical model of the mechanical neuron

This section first characterises the mechanical behaviour of the proposed neuron by examining the effects of the activation-chamber stiffness, the membrane stiffness, the bias membrane term and the input pressures. This characterisation establishes how the physical properties and initial operating conditions of the neuron influence its response. Building on the mechanical characterisation, the analysis then establishes the computational expressiveness of the proposed neuron.

4.1 Mathematical model

The parameters p_{low} and p_{high} define the lower and upper pressure bounds of the system, respectively. These bounds represent the minimum and maximum pressures considered.

Using Eq. (3.12), and singling out the first input, the activation pressure of the neuron is written as

$$p_a = \frac{W_1 p_1 + \sum_{j \neq 1} W_j p_j + W_b p_b + W_0 p_{\text{atm}}}{W_1 + \sum_{j \neq 1} W_j + W_b + W_0}. \quad (4.1)$$

Here, p_j is the j -th input pressure, while W_j is the corresponding mechanical weight. The term $W_b p_b$ acts as a bias, supplied by a bias membrane with weight W_b and bias pressure p_b . The term W_0 represents the compliance of the activation chamber with respect to the atmosphere, and p_{atm} is the atmospheric pressure acting on the chamber housing.

The activation-to-output kernel is represented by a sigmoid function:

$$p_{\text{out}}(p_a) = p_{\text{low}} + \frac{p_{\text{high}} - p_{\text{low}}}{1 + \exp[-k(p_a - p_{\text{th}})]}. \quad (4.2)$$

Here, p_a is the activation pressure, p_{th} is the switching threshold, p_{low} and p_{high} are the limiting output pressures, and k controls the steepness and direction of the transition. A positive k represents the non-inverted configuration, in which the output pressure increases with p_a , whereas a negative k represents the inverted configuration, in which the output pressure decreases as p_a increases.

For simplicity, unless stated otherwise, all subsequent analyses assume that the membrane areas and stiffnesses remain constant. The membranes are therefore modelled as linear elastic elements, such that their restoring forces vary linearly with deformation.

4.2 Mechanical characterisation

For the mechanical characterisation, atmospheric pressure is used as the reference pressure, $p_{\text{atm}} = 0$, and the pressure range considered in the model is bounded by

$$p_{\text{low}} = -100 \text{ kPa}, \quad p_{\text{high}} = 100 \text{ kPa}.$$

4.2.1 Activation-chamber stiffness

The activation-chamber compliance is represented by W_0 . With $p_{\text{atm}} = 0$, Eq. (4.1) becomes

$$p_a = \frac{\sum_j W_j p_j + W_b p_b}{\sum_j W_j + W_b + W_0}. \quad (4.3)$$

Defining the weighted input pressure as

$$p_{\text{in}} = \sum_j W_j p_j + W_b p_b, \quad (4.4)$$

the activation pressure becomes

$$p_a = \frac{1}{\sum_j W_j + W_b + W_0} p_{\text{in}}. \quad (4.5)$$

Let

$$\alpha = \frac{1}{\sum_j W_j + W_b + W_0}. \quad (4.6)$$

Then $p_a = \alpha p_{\text{in}}$. Substituting this relationship into Eq. (4.2) gives

$$p_{\text{out}}(p_{\text{in}}) = p_{\text{low}} + \frac{p_{\text{high}} - p_{\text{low}}}{1 + \exp[-k(\alpha p_{\text{in}} - p_{\text{th}})]}. \quad (4.7)$$

Thus, the activation-chamber compliance is incorporated into the activation function through the scaling factor α . When the function is expressed in terms of the weighted input pressure, this factor modifies the effective gradient of the sigmoid:

$$k_{\text{eff}} = k \alpha. \quad (4.8)$$

Therefore, increasing the chamber compliance reduces the effective gradient of the activation-to-output function. The complete algebraic transformation is provided in Appendix B.

4.2.2 Weight membrane

4.2.2.1 Sensitivity

To examine the influence of the first weight membrane, W_1 is treated as the variable parameter while the remaining pressures and weights are held constant. The activation pressure is

$$p_a = \frac{W_1 p_1 + \sum_{j \neq 1} W_j p_j + W_b p_b + W_0 p_{\text{atm}}}{W_1 + \sum_{j \neq 1} W_j + W_b + W_0}. \quad (4.9)$$

The sensitivity of the activation pressure to the mechanical weight W_1 is (Appendix C)

$$\frac{\partial p_a}{\partial W_1} = \frac{p_1 - p_a}{W_1 + \sum_{j \neq 1} W_j + W_b + W_0}. \quad (4.10)$$

Thus, increasing W_1 shifts the activation pressure towards p_1 . The corresponding sensitivity to the membrane stiffness K_1 , using $W_1 = A_1/K_1$, is

$$\frac{\partial p_a}{\partial K_1} = - \frac{p_1 - p_a}{K_1 + \frac{K_1^2}{A_1} \left(\sum_{j \neq 1} W_j + W_b + W_0 \right)}. \quad (4.11)$$

This expression shows how changes in the membrane stiffness affect the activation pressure. Because the mechanical contribution associated with the membrane is proportional to A_1/K_1 , decreasing K_1 increases the relative contribution of p_1 to p_a .

4.2.2.2 Range

Since $W_1 = A_1/K_1$, the limiting cases are

$$\lim_{K_1 \rightarrow 0} p_a = p_1, \quad (4.12)$$

because $W_1 \rightarrow \infty$ and the contribution of p_1 dominates the weighted average. Conversely,

$$\lim_{K_1 \rightarrow \infty} p_a = \frac{\sum_{j \neq 1} W_j p_j + W_b p_b + W_0 p_{\text{atm}}}{\sum_{j \neq 1} W_j + W_b + W_0}, \quad (4.13)$$

because $W_1 \rightarrow 0$. Thus, a membrane with zero stiffness makes p_a approach p_1 , whereas an infinitely stiff membrane makes its contribution negligible.

4.2.3 Bias membrane term

Starting from Eq. (4.3), and holding all other parameters constant, the quotient rule gives

$$\frac{\partial p_a}{\partial W_b} = \frac{p_b - p_a}{\sum_j W_j + W_b + W_0} = \alpha (p_b - p_a), \quad (4.14)$$

where α is defined in Eq. (4.6). Similarly,

$$\frac{\partial p_a}{\partial p_b} = \frac{W_b}{\sum_j W_j + W_b + W_0} = \alpha W_b. \quad (4.15)$$

When W_b is held constant, $\partial p_a / \partial p_b = \alpha W_b$ remains constant for fixed system parameters. Therefore, p_b acts as a predictable and tuneable bias input. In contrast, holding p_b constant does not make $\partial p_a / \partial W_b = \alpha(p_b - p_a)$ constant, because this sensitivity depends on the changing value of p_a . Consequently, W_b does not act as a static bias.

4.2.4 Input pressure and ANN form

4.2.4.1 Input sensitivity

The input pressures p_j are the externally applied pressure signals that drive the mechanical neuron. Each input acts on a membrane with an associated mechanical weight W_j , and the resulting pressure contributions are combined in the activation chamber. The purpose of this section is to determine how each input pressure influences the activation pressure and to establish the relationship between the mechanical model and the mathematical form of an artificial neural network (ANN).

The sensitivity of the activation pressure to an individual input pressure p_i is (Appendix D)

$$\frac{\partial p_a}{\partial p_i} = \frac{W_i}{\sum_j W_j + W_b + W_0}. \quad (4.16)$$

Using the definition of α ,

$$\frac{\partial p_a}{\partial p_i} = \alpha W_i. \quad (4.17)$$

This result shows that the influence of each input pressure is determined by its corresponding mechanical weight relative to the total effective weight of the system. A larger W_i produces a larger change in p_a for a given change in p_i , while a smaller W_i produces a weaker response.

4.2.4.2 Special case: ANN form

Assuming that W is constant for all membranes, the activation pressure can be written in the same form as an artificial neuron,

$$p_a = \sum_j \left(\frac{\partial p_a}{\partial p_j} \right) p_j + b, \quad (4.18)$$

where b represents the artificial bias contribution. From the derivation in Appendix E, the effective artificial input weights and bias are

$$w_j = \frac{\partial p_a}{\partial p_j} = \alpha W_j \quad (4.19)$$

and

$$b = \alpha W_b p_b = w_b p_b. \quad (4.20)$$

Thus,

$$p_a = \sum_j w_j p_j + b. \quad (4.21)$$

This is the standard pre-activation form of an artificial neuron. In this representation, the pressure p_j corresponds to the neural-network input, the normalised mechanical weight αW_j corresponds to the ANN weight, and the bias-membrane contribution $\alpha W_b p_b$ corresponds to the ANN bias.

The activation-to-output relationship is then applied to the activation pressure,

$$p_{\text{out}} = F(p_a), \quad (4.22)$$

where F is the sigmoid pressure-response function defined in Eq. (4.2). Substituting the equivalent ANN expression for p_a gives

$$p_{\text{out}} = F\left(\sum_j w_j p_j + b\right). \quad (4.23)$$

This establishes that the complete input-to-output kernel of the proposed mechanical neuron is computationally equivalent to that of an artificial neuron when the membrane areas and effective stiffnesses remain constant throughout deformation. In this case, the sensitivity of the activation pressure to each input pressure is constant, allowing the activation pressure to be expressed as a weighted sum of the inputs and a bias term.

4.2.4.3 General form

This, however, represents a special case of the more general mechanical formulation. In general, the membrane stiffness varies with deformation, and the deformation depends on both the activation pressure and the corresponding input pressure. Consequently, the sensitivity of the activation pressure to an input is generally state-dependent,

$$\frac{\partial p_a}{\partial p_i} = \frac{w_i + \sum_j p_j \frac{\partial w_j}{\partial p_i} + \frac{\partial b}{\partial p_i}}{1 - \sum_j p_j \frac{\partial w_j}{\partial p_a} - \frac{\partial b}{\partial p_a}}, \quad (4.24)$$

and the effective mechanical weight can therefore be expressed as a function of the instantaneous state and input. The activation pressure consequently takes the more general form

$$p_a = \sum_j w_j(p_a, \vec{p}, p_b) p_j + b(p_a, \vec{p}, p_b), \quad \vec{p} = [p_1, p_2, \dots, p_N]^\top. \quad (4.25)$$

The neuron is therefore not, in general, equivalent to a conventional fixed-weight artificial neuron. Instead, it constitutes a nonlinear, state-dependent neural computation element.

4.3 Computational expressiveness

The computational expressiveness of the proposed neuron is determined by the space of parameters that can be physically represented by its mechanical architecture. In the constant-stiffness case, where the neuron can be expressed in artificial-neuron form, the input sensitivity is given by Eq. (4.16), while the corresponding bias is

$$b = \frac{W_b p_b}{\sum_j W_j + W_b + W_0}.$$

Consequently, the effective ANN parameters satisfy $\sum_j w_j + w_b + w_0 = 1$. Furthermore, since the underlying membrane areas and stiffnesses cannot be negative, the corresponding mechanical weights are constrained to be positive. The resulting neuron therefore occupies only a restricted subset of the parameter space of a conventional artificial neuron.

The expressiveness is first examined within this artificial-neuron representation, corresponding to the special case of constant stiffness. The analysis then considers the general case, in which membrane stiffness varies with deformation and the effective weights become functions of the activation and input pressures. There, the neuron is treated as a nonlinear, state-dependent computational element, and its expressiveness is considered within this more general parameter space.

4.3.1 Constant-stiffness computational expressiveness

For the constant-stiffness case, the proposed neuron can be represented by the artificial-neuron formulation

$$p_a = \sum_j w_j p_j + w_b p_b + w_0 p_0. \quad (4.26)$$

The parameters w_j , w_b , and p_b are mechanically tuneable, whereas w_0 and p_0 are static parameters representing the compliance of the activation chamber to the atmospheric reference. Consequently, an N -input mechanical neuron has $N + 2$ tuneable parameters: N input weights, the mechanical bias weight w_b , and the tuneable bias pressure p_b . In contrast, a conventional N -input artificial neuron has $N + 1$ parameters, consisting of N weights and a single independent bias term. The additional mechanical parameter arises because the bias contribution is physically represented by the product $w_b p_b$, rather than by a single independently specified parameter.

The pre-activation relation defines a hyperplane in the input-output space, with the weights determining its orientation. Defining $w_j = \tan \theta_j$, each weight determines the inclination θ_j of the hyperplane relative to the corresponding input axis. Since the mechanical parameters from which the weights are formed are non-negative, $w_j \geq 0$, and therefore $\theta_j \geq 0$. Consequently,

$$\frac{\partial p_a}{\partial p_j} = w_j \geq 0, \quad (4.27)$$

meaning that increasing any input pressure cannot decrease the pre-activation pressure. The mechanical neuron is therefore monotonically non-decreasing with respect to each input, and strictly increasing where $w_j > 0$. The additional constraint

$$\sum_j w_j + w_b + w_0 = 1 \quad (4.28)$$

further restricts the representable parameter space by removing one degree of freedom from the weights. This constraint also imposes a geometric property on the resulting hyperplane: it must contain the equal-pressure diagonal, defined by

$$p_1 = p_2 = \dots = p_N = p_b = p_0 = p_a. \quad (4.29)$$

As shown in Appendix F, fixing p_b reduces this diagonal to a specific point through which the hyperplane must pass. Since p_b is itself tuneable, this point can be moved along the equal-pressure diagonal, providing an additional degree of freedom in the position of the hyperplane. Thus, although the constant-stiffness mechanical neuron has $N + 2$ tuneable mechanical parameters, the normalisation constraint leaves $N + 1$ independent degrees of freedom, matching the number of independent parameters of a conventional N -input artificial neuron. However, the resulting parameter space remains restricted by the positivity and normalisation constraints and therefore does not span the full parameter space of a conventional artificial neuron.

4.3.2 General computational expressiveness

For the general case, the normalisation constraint in Eq. (4.28) remains present. Therefore, the derivation presented in Appendix F remains applicable, and the equal-pressure diagonal continues to satisfy the general formulation. For a fixed p_b , the computational surface must consequently pass through the point on the diagonal corresponding to p_b ,

$$(p_1, \dots, p_N, p_a) = (p_b, \dots, p_b, p_b).$$

Thus, p_b continues to determine the point at which the computational surface intersects the equal-pressure diagonal, while the state-dependent weights determine the local shape and orientation of the surface around this point.

The positivity constraint on the mechanical parameters also remains. Although the membrane stiffness is no longer constant, its value remains physically positive,

$$K_i(\delta_i) > 0,$$

for all admissible deformations δ_i . Consequently, the effective mechanical weights remain within the positive domain. However, unlike the constant-stiffness case, $K_i(\delta_i)$ is generally neither constant nor necessarily monotonic with deformation. The effective weights therefore vary with the mechanical state, and the resulting input-to-output mapping is not necessarily monotonically increasing with each input. The general neuron consequently remains constrained to the physically positive mechanical parameter domain while allowing a nonlinear, state-dependent computational surface within that domain.

5 Conclusion

This article established the theoretical, structural, and mathematical framework for a mechanically feasible neuroplastic neuron designed for soft robotic applications. By translating biological neural processes into fluidic mechanisms, the proposed design demonstrates how pressure-driven inputs can be integrated through membrane interactions into a unified activation state without consuming or destroying the driving input signals. Furthermore, physical mechanisms for mechanical weight tuning, namely deformation limiting and intermediate-fluid bulk-modulus regulation, provide a proposed physical medium for post-fabrication reconfigurability and future neuroplasticity.

Mathematically, the proposed architecture maps directly onto the weighted-sum and sigmoidal activation forms of a conventional artificial neural network under linear, constant-stiffness assumptions. When membrane stiffness varies with deformation, the system generalises into a nonlinear, state-dependent neural computation element. While physical constraints impose positivity and normalisation bounds on the reachable parameter space, the architecture retains the expressiveness to serve as a building block for physical neural computation.

Future research directions include the physical fabrication and empirical characterisation of individual units to analyse real-world fluid dynamics and mechanical response boundaries. Beyond single-unit mechanics, exploring the integration of these neurons into multi-layered architectures presents a critical pathway toward fully autonomous, self-learning mechanical neural networks capable of physical adaptation in soft robotic systems. Central to this progression is establishing fully mechanical parameter-update mechanisms, enabling synaptic weights to adapt locally without relying on external domain interventions.

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Appendix A Intermediate-fluid bulk-modulus tuning

A.1 Definitions and sign convention

The force-transfer-limiting mechanism consists of two flexible membranes separated by a confined intermediate fluid. The input pressure p_i acts on membrane 1, the activation-chamber pressure p_a acts on membrane 2, and the intermediate fluid develops the weight pressure p_w .

The effective areas of membranes 1 and 2 are denoted by A_1 and A_2 , respectively. Their volumetric deformations into the adjacent chambers are denoted by ΔV_1 and ΔV_2 , with corresponding volumetric stiffnesses K_1 and K_2 . The deformation relevant to the activation response is the deformation of membrane 2,

$$\Delta V_j = \Delta V_2. \quad (\text{A.1})$$

The effective stiffness of the complete force-transfer mechanism is defined as

$$K_j = \frac{F_j}{\Delta V_j}, \quad (\text{A.2})$$

where F_j is the effective force producing deformation into the activation chamber.

A.2 Individual membrane force balances

For membrane 1, the pressure difference between the input chamber and intermediate chamber produces the force

$$F_1 = (p_i - p_w) A_1. \quad (\text{A.3})$$

Using the volumetric stiffness relation $F_1 = K_1 \Delta V_1$, the deformation of membrane 1 is

$$\Delta V_1 = \frac{(p_i - p_w) A_1}{K_1}. \quad (\text{A.4})$$

For membrane 2, the pressure difference between the intermediate chamber and activation chamber produces the force

$$F_2 = (p_w - p_a) A_2. \quad (\text{A.5})$$

Similarly, $F_2 = K_2 \Delta V_2$, and therefore

$$\Delta V_j = \Delta V_2 = \frac{(p_w - p_a) A_2}{K_2}. \quad (\text{A.6})$$

A.3 Intermediate-fluid compressibility

Let V_p denote the initial volume of the intermediate fluid and B its bulk modulus. For small changes, the bulk modulus relates the pressure change to the relative volume change according to

$$\Delta p_w = -B \frac{\Delta V_p}{V_p}. \quad (\text{A.7})$$

Using the pressure increment relative to the initial weight pressure $p_{w,0}$,

$$p_w - p_{w,0} = -B \frac{\Delta V_p}{V_p}. \quad (\text{A.8})$$

With the adopted deformation convention, the change in intermediate-fluid volume is

$$\Delta V_p = \Delta V_2 - \Delta V_1. \quad (\text{A.9})$$

Thus,

$$p_w - p_{w,0} = -\frac{B}{V_p} (\Delta V_2 - \Delta V_1). \quad (\text{A.10})$$

Substituting the individual membrane deformations gives

$$p_w - p_{w,0} = -\frac{B}{V_p} \left[\frac{(p_w - p_a) A_2}{K_2} - \frac{(p_i - p_w) A_1}{K_1} \right]. \quad (\text{A.11})$$

This equation determines the weight pressure p_w as a function of the input pressure, activation pressure, membrane properties, and intermediate-fluid bulk modulus.

A.4 Effective stiffness

The effective force associated with the complete two-membrane mechanism is written as

$$F_j = F_1 + F_2 = (p_i - p_w) A_1 + (p_w - p_a) A_2. \quad (\text{A.12})$$

The effective stiffness is therefore, with $\Delta V_j = \Delta V_2$,

$$K_j = \frac{F_j}{\Delta V_j} = \frac{(p_i - p_w) A_1 + (p_w - p_a) A_2}{\Delta V_2}. \quad (\text{A.13})$$

Substituting the expression for ΔV_2 gives

$$K_j = \frac{(p_i - p_w) A_1 + (p_w - p_a) A_2}{\frac{(p_w - p_a) A_2}{K_2}} = K_2 \frac{(p_i - p_w) A_1 + (p_w - p_a) A_2}{(p_w - p_a) A_2}. \quad (\text{A.14})$$

Because p_w depends on the bulk modulus B , the effective stiffness is also a function of B , $K_j = K_j(B)$, and the corresponding mechanical weight is

$$W_j(B) = \frac{A_j}{K_j(B)}. \quad (\text{A.15})$$

A.5 Limiting cases

A.5.1 Completely compressible intermediate fluid

For a completely compressible intermediate fluid, $B = 0$. The intermediate fluid provides no resistance to volumetric deformation and cannot sustain a pressure increase. Consequently, the force-transfer mechanism does not produce a finite activation-side deformation associated with the input force, $\Delta V_j \rightarrow 0$. Therefore,

$$K_j(B = 0) = \frac{F_j}{\Delta V_j} \rightarrow \infty. \quad (\text{A.16})$$

Thus, the completely compressible limit corresponds to an effectively infinitely stiff force-transfer mechanism.

A.5.2 Completely incompressible intermediate fluid

For a completely incompressible intermediate fluid, $B \rightarrow \infty$. The intermediate-fluid volume cannot change, so $\Delta V_p \rightarrow 0$. Under the adopted sign convention, this requires the deformation of the two membranes to be equal:

$$\Delta V_1 = \Delta V_2 = \Delta V_j. \quad (\text{A.17})$$

The two membrane stiffnesses therefore act together in the fully coupled limit:

$$F_j = K_1 \Delta V_j + K_2 \Delta V_j = (K_1 + K_2) \Delta V_j, \quad (\text{A.18})$$

and the effective stiffness becomes

$$K_j(B \rightarrow \infty) = \frac{F_j}{\Delta V_j} = K_1 + K_2. \quad (\text{A.19})$$

The limiting behaviour of the force-transfer mechanism is therefore

$$K_j(0) \rightarrow \infty, \quad K_j(\infty) = K_1 + K_2. \quad (\text{A.20})$$

The intermediate-fluid bulk modulus consequently provides a tuneable parameter that controls the effective stiffness of the two-membrane system and, through $W_j(B) = A_j/K_j(B)$, the corresponding mechanical weight. Bulk modulus is conventionally defined as the ratio between pressure change and volumetric strain, with the sign determined by the selected convention for compression and volume change.

Appendix B Activation-chamber compliance

Starting with the activation model with $p_{\text{atm}} = 0$,

$$p_a = \frac{\sum_j W_j p_j + W_b p_b}{\sum_j W_j + W_b + W_0}, \quad (\text{B.1})$$

define $p_{\text{in}} = \sum_j W_j p_j + W_b p_b$, so that

$$p_a = \left(\frac{1}{\sum_j W_j + W_b + W_0} \right) p_{\text{in}} = \alpha p_{\text{in}}, \quad \alpha = \frac{1}{\sum_j W_j + W_b + W_0}. \quad (\text{B.2})$$

The activation-to-output function is

$$p_{\text{out}}(p_a) = p_{\text{low}} + \frac{p_{\text{high}} - p_{\text{low}}}{1 + \exp[-k(p_a - p_{\text{th}})]}. \quad (\text{B.3})$$

Substituting $p_a = \alpha p_{\text{in}}$ yields

$$p_{\text{out}}(p_{\text{in}}) = p_{\text{low}} + \frac{p_{\text{high}} - p_{\text{low}}}{1 + \exp[-k(\alpha p_{\text{in}} - p_{\text{th}})]}. \quad (\text{B.4})$$

The exponent can be rearranged as

$$-k(\alpha p_{\text{in}} - p_{\text{th}}) = -k\alpha \left(p_{\text{in}} - \frac{p_{\text{th}}}{\alpha} \right). \quad (\text{B.5})$$

Consequently,

$$p_{\text{out}}(p_{\text{in}}) = p_{\text{low}} + \frac{p_{\text{high}} - p_{\text{low}}}{1 + \exp[-k_{\text{eff}}(p_{\text{in}} - p_{\text{th,eff}})]}, \quad (\text{B.6})$$

where

$$k_{\text{eff}} = k\alpha = \frac{k}{\sum_j W_j + W_b + W_0} \quad \text{and} \quad p_{\text{th,eff}} = \frac{p_{\text{th}}}{\alpha} = p_{\text{th}} \left(\sum_j W_j + W_b + W_0 \right). \quad (\text{B.7})$$

The activation-chamber compliance therefore acts as a multiplicative scaling factor on the weighted input pressure. Consequently, it changes the gradient of the sigmoid response: increasing the chamber compliance reduces the effective gradient, while decreasing the compliance produces a steeper response.

Appendix C Weight-membrane sensitivity

Define

$$N = W_1 p_1 + \sum_{j \neq 1} W_j p_j + W_b p_b + W_0 p_{\text{atm}}, \quad D = W_1 + \sum_{j \neq 1} W_j + W_b + W_0, \quad (\text{C.1})$$

so that the activation pressure is $p_a = N/D$. Differentiating with respect to W_1 , while treating all other pressures and weights as constant, gives

$$\frac{\partial p_a}{\partial W_1} = \frac{\frac{\partial N}{\partial W_1} D - N \frac{\partial D}{\partial W_1}}{D^2}. \quad (\text{C.2})$$

Since $\partial N / \partial W_1 = p_1$ and $\partial D / \partial W_1 = 1$, it follows that

$$\frac{\partial p_a}{\partial W_1} = \frac{p_1 D - N}{D^2}. \quad (\text{C.3})$$

Using $N = p_a D$,

$$\frac{\partial p_a}{\partial W_1} = \frac{p_1 D - p_a D}{D^2}, \quad (\text{C.4})$$

which simplifies to

$$\boxed{\frac{\partial p_a}{\partial W_1} = \frac{p_1 - p_a}{D}}. \quad (\text{C.5})$$

The mechanical weight is related to the membrane stiffness by $W_1 = A_1 / K_1$. Therefore,

$$\frac{\partial W_1}{\partial K_1} = -\frac{A_1}{K_1^2}. \quad (\text{C.6})$$

Applying the chain rule,

$$\frac{\partial p_a}{\partial K_1} = \frac{\partial p_a}{\partial W_1} \frac{\partial W_1}{\partial K_1} = -\frac{A_1}{K_1^2} \frac{p_1 - p_a}{D}. \quad (\text{C.7})$$

Hence,

$$\boxed{\frac{\partial p_a}{\partial K_1} = -\frac{A_1 (p_1 - p_a)}{K_1^2 (W_1 + \sum_{j \neq 1} W_j + W_b + W_0)}}. \quad (\text{C.8})$$

Appendix D Input-pressure sensitivity

With $p_{\text{atm}} = 0$, the activation pressure is

$$p_a = \frac{\sum_j W_j p_j + W_b p_b}{\sum_j W_j + W_b + W_0}. \quad (\text{D.1})$$

The denominator is independent of the input pressures p_i . Differentiating with respect to p_i therefore gives

$$\frac{\partial p_a}{\partial p_i} = \frac{\partial (\sum_j W_j p_j + W_b p_b) / \partial p_i}{\sum_j W_j + W_b + W_0}. \quad (\text{D.2})$$

Since

$$\frac{\partial}{\partial p_i} \left(\sum_j W_j p_j \right) = W_i \quad (\text{D.3})$$

and the bias term $W_b p_b$ is independent of p_i ,

$$\frac{\partial p_a}{\partial p_i} = \frac{W_i}{\sum_j W_j + W_b + W_0}. \quad (\text{D.4})$$

Using α ,

$$\boxed{\frac{\partial p_a}{\partial p_i} = \alpha W_i}. \quad (\text{D.5})$$

Appendix E Equivalent ANN weights and bias

With $p_{\text{atm}} = 0$, the activation pressure is

$$p_a = \alpha \left(\sum_j W_j p_j + W_b p_b \right), \quad (\text{E.1})$$

where α is defined in Eq. (4.6). Differentiating with respect to an input pressure p_j gives

$$\frac{\partial p_a}{\partial p_j} = \alpha W_j. \quad (\text{E.2})$$

The activation pressure can therefore be expressed in the artificial-neural-network form

$$p_a = \sum_j \left(\frac{\partial p_a}{\partial p_j} \right) p_j + b. \quad (\text{E.3})$$

Substituting the derived input sensitivities gives

$$p_a = \sum_j \alpha W_j p_j + b = \alpha \sum_j W_j p_j + b. \quad (\text{E.4})$$

Equating this expression with the original activation-pressure equation gives

$$\alpha \sum_j W_j p_j + b = \alpha \left(\sum_j W_j p_j + W_b p_b \right). \quad (\text{E.5})$$

After cancelling the common input-pressure term,

$$b = \alpha W_b p_b = \frac{W_b p_b}{\sum_j W_j + W_b + W_0}. \quad (\text{E.6})$$

Consequently, the activation pressure has the equivalent artificial-neural-network form

$$p_a = \sum_j w_j p_j + b, \quad (\text{E.7})$$

where the effective ANN weights are

$$w_j = \frac{\partial p_a}{\partial p_j} = \alpha W_j = \frac{W_j}{\sum_k W_k + W_b + W_0}. \quad (\text{E.8})$$

Applying the activation-to-output function then gives

$$p_{\text{out}} = F(p_a) = F \left(\sum_j w_j p_j + b \right). \quad (\text{E.9})$$

Appendix F Equal-pressure hyperplane

Consider the pre-activation relation

$$p_a = \sum_j w_j p_j + w_b p_b + w_0 p_0 \quad (\text{F.1})$$

subject to

$$\sum_j w_j + w_b + w_0 = 1. \quad (\text{F.2})$$

Let all input and reference pressures be equal to some arbitrary pressure p , $p_j = p_b = p_0 = p$. Substituting into the pre-activation equation gives

$$p_a = \sum_j w_j p + w_b p + w_0 p = \left(\sum_j w_j + w_b + w_0 \right) p. \quad (\text{F.3})$$

Using the normalisation constraint, $p_a = p$. Therefore

$$\boxed{p_1 = p_2 = \dots = p_N = p_b = p_0 = p_a} \quad (\text{F.4})$$

satisfies the pre-activation equation for any value of p . Hence, the hyperplane necessarily contains this one-dimensional equal-pressure diagonal.

Thus, the normalisation constraint does not merely restrict the magnitudes of the weights; it geometrically constrains the family of representable hyperplanes to those passing through the equal-pressure diagonal. For a fixed value of p_b , the hyperplane is constrained to pass through the corresponding point on the equal-pressure diagonal,

$$(p_1, p_2, \dots, p_N, p_a) = (p_b, p_b, \dots, p_b, p_b). \quad (\text{F.5})$$

Since p_b is a mechanically tuneable parameter, this point can be moved along the diagonal, providing control over the position of the hyperplane. The input weights w_j , meanwhile, determine its orientation, with w_b determined by the normalisation constraint. Thus, the mechanical parameters provide control over both the orientation and position of the hyperplane, although only within the restrictions imposed by the non-negative mechanical parameters and the normalisation constraint.